Amazon Sales Exploratory Data Analysis (EDA)

Internship Project Report

Submitted By

Samruddhi Bhowood

Submitted To

CodeAlpha Internship

Date

10 May 2026

1. Introduction

Exploratory Data Analysis (EDA) is an important step in the data analysis process used to understand datasets, identify trends, detect anomalies, and generate meaningful insights.

This project focuses on performing Exploratory Data Analysis on an Amazon sales dataset obtained from Kaggle. The dataset contains product information such as product names, categories, prices, discounts, ratings, and review counts.

The main objective of this project is to analyze customer behavior, pricing patterns, product ratings, and discount trends using Python data analysis libraries.

---

2. Project Objective

The objectives of this project are:

- To understand the structure of the dataset
- To clean and preprocess the data
- To identify missing values and anomalies
- To analyze product ratings and pricing patterns
- To detect outliers and inconsistencies
- To discover trends and insights from Amazon product data
- To visualize data using charts and graphs

---

3. Dataset Description

The dataset used in this project was downloaded from Kaggle.

Dataset Features

The dataset contains the following important columns:

- Product_name
- category
- actual_price
- discounted_price
- discount_percentage
- rating
- rating_count

Dataset Source

Kaggle Amazon Sales Dataset

---

4. Technologies Used

Technology	Purpose
Python	Data Analysis
Pandas	Data Cleaning and Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
VS Code	Development Environment

---

5. Problem Statement

The objective of this project is to perform Exploratory Data Analysis (EDA) on Amazon product data and:

- Ask meaningful questions before analysis
- Explore the dataset structure and data types
- Identify trends, patterns, and anomalies
- Validate assumptions using visualization and statistics
- Detect potential data quality issues
- Generate business insights from the dataset

---

6. Meaningful Questions Before Analysis

Before starting the analysis, the following questions were considered:

1. Which product category has the highest average rating?
2. What is the distribution of product ratings?
3. Do higher discounts improve product ratings?
4. Which categories dominate the dataset?
5. Are there any outliers in product prices?
6. Which products receive the highest discounts?

7. Is there any relationship between product price and ratings?
8. Which products have the highest number of reviews?

---

7. Data Cleaning and Preprocessing

Several preprocessing steps were performed before analysis.

Cleaning Steps Performed

- Removed currency symbols from price columns
- Removed commas from numerical values
- Converted object datatypes into float datatype
- Handled missing values
- Removed duplicate records
- Fixed encoding issues in price columns
- Converted rating and review columns into numeric datatype

Data Issues Identified

During the cleaning process, several data quality issues were identified:

- Price columns contained currency symbols and commas
- Encoding issues were present in some rows
- Missing values existed in rating-related columns
- Several outliers were detected in product pricing
- The dataset was heavily dominated by electronics-related products

---

8. Exploratory Data Analysis

8.1 Distribution of Product Ratings

Observation

Most product ratings are concentrated between 4.0 and 4.5.

Insight

This indicates that the majority of products in the dataset receive positive customer feedback and generally maintain good ratings.

8.2 Distribution of Discount Percentage

Observation

Many products offer moderate to high discounts.

Insight

Amazon products frequently use discounts as a pricing strategy to attract customers.

8.3 Top Product Categories

Observation

Electronics and accessories categories dominate the dataset.

Insight

The dataset mainly focuses on electronic and technology-related products.

8.4 Product Price Outliers

Observation

Several products have extremely high prices compared to the majority of products.

Insight

These products represent premium or luxury items and create significant outliers in the dataset.

8.5 Correlation Analysis

A correlation analysis was performed between discount percentage and product ratings.

Correlation Value

-0.155

Observation

A weak negative correlation exists between discount percentage and ratings.

Insight

Higher discounts do not significantly improve customer ratings.

---

9. Visualization Techniques Used

The following visualizations were used during analysis:

- Histogram
- Boxplot
- Bar Chart
- Correlation Heatmap
- Distribution Plot

These visualizations helped identify trends, patterns, distributions, and anomalies in the dataset.

---

10. Key Insights

The following important insights were discovered during the EDA process:

1. Most products are rated between 4.0 and 4.5.
2. Electronics-related categories dominate the dataset.
3. Several products offer extremely high discounts.
4. Significant price outliers exist in the dataset.
5. Products with higher discounts do not necessarily receive better ratings.
6. Highly reviewed products generally maintain stable ratings.
7. Customer ratings are mostly positive across categories.

---

11. Challenges Faced

During the project, the following challenges were encountered:

- Encoding issues in currency symbols
- Converting price columns into numeric datatype
- Handling missing values
- Managing outliers in pricing data

- Understanding relationships between variables

These challenges were resolved through preprocessing and data cleaning techniques.

---

12. Conclusion

The Exploratory Data Analysis process successfully helped in understanding the Amazon products dataset and identifying meaningful patterns and insights.

The project revealed that most products receive positive customer ratings and that electronics dominate the dataset. Outliers in product pricing were also identified. Correlation analysis showed that discounts have only a weak relationship with customer ratings.

This project improved understanding of data cleaning, visualization, statistical analysis, and insight generation using Python.

---

13. Future Scope

This project can be further extended by:

- Building a recommendation system
- Performing sentiment analysis on reviews
- Predicting product ratings using machine learning
- Creating interactive dashboards using Power BI or Tableau
- Developing sales forecasting models

---

14. References

- Kaggle Dataset
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Python Official Documentation

---

16. Final Outcome

This project successfully demonstrated the practical implementation of Exploratory Data Analysis using Python libraries. The project helped in identifying trends, detecting anomalies, validating assumptions, and generating business insights from Amazon product data.

The project also strengthened practical skills in:

- Data Cleaning
- Data Visualization
- Statistical Analysis
- Python Programming
- Insight Generation
- GitHub Project Management